Chapter 13. Identifiable Gaussian Linear SEMs

Graphical models, Lecture 6 (equal error variances, conditional-variance ordering, ANM, the US algorithm)

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1 In General Gaussian SEMs, Directions Are Not Identified

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

These give the same CIs and Gaussian distribution (Ch. 9).

General Gaussian SEM: $\epsilon_j \sim N(0, \sigma_j^2)$	Equal-variance Gaussian SEM: $\epsilon_j \sim N(0, \sigma^2)$
if each variable may have its own error variance, several directions are compatible with the same distribution result: CPDAG	if all structural errors are restricted to the same variance, the residual-variance structure differs by direction result: DAG

Even for Gaussian data, an assumption on the *error variances* identifies the full DAG.

2 The Equal-Variance Gaussian SEM

Definition: equal-variance Gaussian SEM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad \epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

$$X = BX + \epsilon, \qquad \Sigma = \sigma^2 (I - B)^{-1} (I - B)^{-\top}.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering, B is strictly lower triangular
independent errors	no hidden common cause among the analyzed variables
common error variance	all structural errors have the same σ^2
nonzero edges	coefficients of the edges in the graph are not 0

3 Two Variables: The Equal-Variance Assumption Picks the Direction

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2).$$

Reverse $Y \to X$: $Y = \epsilon'_Y$, $X = cY + \eta$ with

$$\mathrm{Var}(\epsilon'_Y) = (1 + b^2)\sigma^2, \qquad \mathrm{Var}(\eta) = \frac{\sigma^2}{1 + b^2}.$$

Proof [Supplement]. $\mathrm{Var}Y = (1 + b^2)\sigma^2$, $c = b/(1 + b^2)$, $\mathrm{Var}(X \mid Y) = \sigma^2 - \dfrac{b^2\sigma^4}{(1 + b^2)\sigma^2} = \dfrac{\sigma^2}{1 + b^2}$. ■

Equal only if $(1 + b^2)^2 = 1$, i.e. $b = 0$. **Only the true direction has a common error variance.**

4 Conditional Variances Reveal the Causal Order

Motivation. With equal noise variances, a source has the smallest (conditional) variance; repeating gives the causal order.

4.1 The key lemma

Theorem: conditional-variance ordering

In an equal-variance Gaussian SEM, let S be an *ancestral* set (it contains all ancestors of each of its members, e.g. the first few variables of a causal order) and $j \notin S$. Then

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \quad \text{if } \mathrm{pa}(j) \subseteq S, \qquad \mathrm{Var}(X_j \mid X_S) \geq \sigma^2(1 + b_{jk}^2) > \sigma^2 \quad \text{otherwise,}$$

for a suitable parent $k \notin S$ of j.

Proof [Supplement]. $X = A\epsilon$, $A = (I - B)^{-1}$, $a_{jj} = 1$, a_{jl} the total effect (0 unless $l \in \mathrm{an}(j) \cup \{j\}$). S ancestral $\Rightarrow$ X_S and $(\epsilon_l)_{l \in S}$ determine each other (triangular system). So in $X_j = \sum_{l \in S} a_{jl}\epsilon_l + \sum_{l \notin S} a_{jl}\epsilon_l$ the first sum is fixed by X_S and the second is independent of it:

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \sum_{l \notin S} a_{jl}^2 = \sigma^2\Big(1 + \sum_{l \notin S,\, l \neq j} a_{jl}^2\Big).$$

If $\mathrm{pa}(j) \subseteq S$, all ancestors of j lie in S and the sum is empty. Otherwise take the parent $k \notin S$ *latest* in a causal order. Another directed path $k \to \cdots \to m \to j$ would give a parent $m \notin S$ later than k, a contradiction; so $a_{jk} = b_{jk} \neq 0$. ■

Given the ordered variables, sources of the rest have conditional variance σ^2, others more:

Theorem: identifiability (Peters and Bühlmann, 2014)

The DAG and B of an equal-variance Gaussian SEM are identified from Σ. A causal order is obtained top-down: $\pi_1 = \arg\min_j \mathrm{Var}(X_j)$, and $\pi_m = \arg\min_{j \notin S} \mathrm{Var}(X_j \mid X_S)$ with $S = \{\pi_1, \ldots, \pi_{m-1}\}$.

4.2 A worked example

Example: three variables

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3, \qquad \epsilon_1, \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 1).$$

Step	Variance comparison	Choice
1	$\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = 1.64$, $\mathrm{Var}(X_3) = 1.8036$	X_1
2	$\mathrm{Var}(X_2 \mid X_1) = 1$, $\mathrm{Var}(X_3 \mid X_1) = 1.49$	X_2
3	the last remaining variable	X_3

$\mathrm{Var}(X_3) = 0.49 \times 1.64 + 1$, $\mathrm{Var}(X_3 \mid X_1) = 0.49\,\mathrm{Var}(X_2 \mid X_1) + 1 = 1.49$. Order $X_1 < X_2 < X_3$.

4.3 After the order: select parents and edges

With order $X_1 < X_2 < X_3$:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

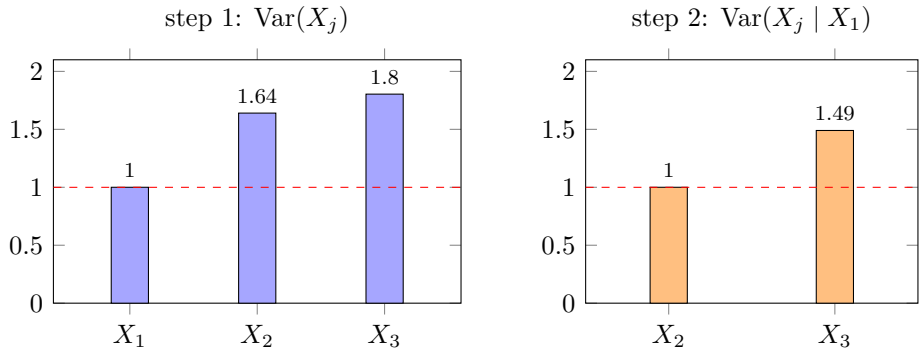


Figure 1: Top-down ordering ($\sigma^2 = 1$, red dashed): the variable at the common error variance is selected, first X_1, then X_2.

$\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{31} \approx 0$, $\hat{\beta}_{32} \approx 0.7$:

$$X_1 \to X_2 \to X_3.$$

5 Equal Variance Does Not Mean Equal Observed Variances

Misconception	Correct meaning
all X_j must have the same variance	variation transmitted from the parents makes observed variances differ (as in the example)
standardizing all variables gives equal variance	standardization rescales the structural errors too, so it does not guarantee the equal-variance assumption — it typically destroys it
similar fitted residual variances prove the assumption	only a diagnostic compatible with the chosen DAG; it does not prove the true direction

Natural for same-kind, same-unit variables with similar measurement error (e.g. expression levels).

Remark: scale dependence and varsortability

The assumption is not scale-invariant (Exercise 13.2). Simulated benchmarks often have variances increasing along the order ("varsortability"), flattering variance-based methods; real data may lack this.

6 Additive Noise Models and the US Algorithm

6.1 The equal-variance Gaussian SEM is a special ANM

$$X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j, \qquad \epsilon_1, \dots, \epsilon_p \text{ independent.}$$

With S the already selected variables,

$$U_j(S) = \mathbb{E}\{\mathrm{Var}(X_j \mid X_S)\}.$$

For ancestral S, $j \notin S$ (Exercise 13.3):

$$U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}].$$

Conditioning set S	Conditional variance
all parents are in S	$U_j(S) = \sigma_j^2$
some parent not yet in S	uncertainty created by the parents is added

6.2 ANMs are identified by an ordering of conditional variances

With true order π and $S_{m-1} = \{\pi_1, \dots, \pi_{m-1}\}$, if

$$U_{\pi_m}(S_{m-1}) < U_k(S_{m-1}), \qquad k \notin S_{m-1} \cup \{\pi_m\},$$

the variables are identified one by one.

Equal-variance Gaussian SEM	General ANM
$\sigma_1^2 = \cdots = \sigma_p^2$ a variable whose parents are all included has no extra uncertainty, hence the smallest conditional variance (lemma above)	$\sigma_1^2, \dots, \sigma_p^2$ may differ identified as long as the order of conditional variances, combining error variance and parent-induced uncertainty, is maintained

This needs neither linearity nor Gaussianity, only the *correct conditional-variance order* (other ANM results use residual independence, as in LiNGAM).

6.3 The US algorithm

Algorithm: US — order by conditional variances, then parents

Step 1: estimate the causal order. Let S be the selected set (initially $\varnothing$).

1. Regress each X_j, $j \notin S$, on X_S (flexibly, if f_j may be nonlinear).
2. Compute the residual variance $\widehat{U}_j(S)$.
3. Select the variable with the smallest value, $\hat{\pi}_m = \arg\min_{j \notin S} \widehat{U}_j(S)$.

Add the selected variable to S and repeat.
Step 2: select the parents. The candidate parents of each variable are the variables before it in the order. For each candidate, test conditional dependence given the other preceding variables:

$$X_{\pi_m} \not\perp\!\!\!\perp X_{\pi_j} \mid X_{\{\pi_1,\dots,\pi_{m-1}\}\setminus\{\pi_j\}}.$$

If conditional dependence remains, add the edge $\pi_j \to \pi_m$.

In a correct order, preceding non-parents are non-descendants, so Markov makes them conditionally independent and faithfulness keeps parents dependent.

7 Summary

Model	Extra assumption	Clue for the direction
LiNGAM	linear relations with independent non-Gaussian errors	independence of the cause and the regression residual
equal-variance Gaussian SEM	common variance of independent Gaussian errors	order of conditional variances starting from the source
ANM and US	an additive noise model that keeps the conditional-variance order	conditional variances find the order; conditional independence selects the parents

All still need acyclicity, no hidden confounding and causal minimality.

Key points.

- General Gaussian SEMs identify only the equivalence class; a common error variance identifies the DAG.
- Sources have the smallest conditional variance; regression then selects edges.
- Check scales, measurement error and confounding.

8 Selected exercises

Motivation. Bottom-up ordering via Ω, unit changes that reverse the direction, and the ANM variance decomposition.

Exercise 13.1 (Bottom-up ordering from the precision matrix)

In an equal-variance Gaussian SEM with nonzero edge coefficients, let $\Omega = \Sigma^{-1}$.
(a) Show that $\Omega_{jj} = \{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}/\sigma^2$, where $\mathrm{ch}(j)$ is the set of children of j. Deduce that $\mathrm{Var}(X_j \mid X_{-j}) = \sigma^2$ if j is a sink (has no children) and $< \sigma^2$ otherwise.
(b) Show that after deleting a sink, the remaining variables again follow an equal-variance Gaussian SEM on the induced subgraph, so that repeatedly removing $\arg\min_j \Omega_{jj}$ yields a reversed causal order.

Solution [Supplement].

(a)

Step 1. $\Omega = (I - B)^\top (I - B)/\sigma^2$ (Ch. 9), so $\Omega_{jj} = \sum_l (I - B)_{lj}^2/\sigma^2$: $l = j$ gives 1, $l \neq j$ gives b_{lj}^2, nonzero iff $j \to l$.

Step 2. $\mathrm{Var}(X_j \mid X_{-j}) = 1/\Omega_{jj} = \sigma^2/\{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}$ (Ch. 8), $= \sigma^2$ iff $\mathrm{ch}(j) = \varnothing$.

(b)

Step 3. A sink appears in no other equation, so the other equations with their $N(0, \sigma^2)$ errors form an equal-variance SEM on the DAG minus j and determine the law of X_{-j}.

Step 4. Sinks have $\Omega_{jj} = 1/\sigma^2$, non-sinks more, so the argmin is a sink; repeating lists variables in reverse causal order. ■

Exercise 13.2 (A change of units can reverse the direction)

Let $X = \epsilon_X$, $Y = bX + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2)$, $b \neq 0$, and $\tilde{Y} = sY$ with $s > 0$.
(a) Show that $(X, \tilde{Y})$ follows the Gaussian SEM $X \to \tilde{Y}$ with error variances σ^2 and $s^2\sigma^2$, which is equal-variance only for $s = 1$.
(b) Show that the top-down rule applied to $(X, \tilde{Y})$ selects $\tilde{Y}$ first iff $s^2 < 1/(1 + b^2)$.
(c) Show that for $s = 1/(1 + b^2)$ the pair $(X, \tilde{Y})$ is *exactly* an equal-variance Gaussian SEM with $\tilde{Y} \to X$.

Solution [Supplement].

(a) $\tilde{Y} = sbX + s\epsilon_Y$, $s\epsilon_Y \sim N(0, s^2\sigma^2) \perp\!\!\!\perp X$.

(b) $\mathrm{Var}X = \sigma^2$, $\mathrm{Var}\tilde{Y} = s^2(1 + b^2)\sigma^2$; $\tilde{Y}$ first iff $s^2(1 + b^2) < 1$.

(c)

Step 1. The reverse factorization is $Y = \epsilon'_Y$, $X = cY + \eta$, $\mathrm{Var}\eta = \sigma^2/(1 + b^2)$, $\eta \perp\!\!\!\perp Y$; so $X = (c/s)\tilde{Y} + \eta$ with $\eta \perp\!\!\!\perp \tilde{Y}$.

Step 2. Error variances $s^2(1 + b^2)\sigma^2$ and $\sigma^2/(1 + b^2)$ coincide at $s = 1/(1 + b^2)$; with Gaussianity this is an equal-variance SEM $\tilde{Y} \to X$.

The method then confidently outputs the wrong direction. ■

Exercise 13.3 (Conditional variances in an additive noise model)

In an ANM $X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j$ with independent errors, $\mathbb{E}\epsilon_j = 0$, $\mathrm{Var}\epsilon_j = \sigma_j^2$ and $\mathbb{E}f_j(X_{\mathrm{pa}(j)})^2 < \infty$, let S be an ancestral set and $j \notin S$.
(a) Show that ϵ_j is independent of $(X_S, X_{\mathrm{pa}(j)})$.
(b) Show that $\mathrm{Var}(X_j \mid X_S) = \sigma_j^2 + \mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}$ almost surely; hence $U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}]$, which equals σ_j^2 when $\mathrm{pa}(j) \subseteq S$.

Solution [Supplement].

(a)

Step 1. By induction along a causal order, X_k is a function of $(\epsilon_l)_{l \in \mathrm{an}(k) \cup \{k\}}$.

Step 2. For $k \in S$, $\mathrm{an}(k) \cup \{k\} \subseteq S \not\ni j$; for $k \in \mathrm{pa}(j)$, $\mathrm{an}(k) \cup \{k\} \subseteq \mathrm{an}(j) \not\ni j$. So $(X_S, X_{\mathrm{pa}(j)})$ is a function of $(\epsilon_l)_{l \neq j} \perp\!\!\!\perp \epsilon_j$.

(b)

Step 3. With $F = f_j(X_{\mathrm{pa}(j)})$, $\bar{F} = \mathbb{E}(F \mid X_S)$, (a) gives $\mathbb{E}(X_j \mid X_S) = \bar{F}$, so

$$\mathrm{Var}(X_j \mid X_S) = \mathbb{E}\{(F - \bar{F})^2 \mid X_S\} + 2\,\mathbb{E}\{(F - \bar{F})\epsilon_j \mid X_S\} + \mathbb{E}(\epsilon_j^2 \mid X_S).$$

Step 4. $\epsilon_j \perp\!\!\!\perp (F, X_S)$: the cross term is $\mathbb{E}\epsilon_j \cdot \mathbb{E}(F - \bar{F} \mid X_S) = 0$ and the last is σ_j^2; the first is $\mathrm{Var}(F \mid X_S)$. Take expectations for $U_j(S)$.

Step 5. If $\mathrm{pa}(j) \subseteq S$, F is X_S-measurable, so $\mathrm{Var}(F \mid X_S) = 0$ and $U_j(S) = \sigma_j^2$. ■